

My goal for this final project was reflective of something that I, and many other recent college graduates have been experiencing these past couple of years: getting a job. What used to be, a few decades ago at least, a far more straightforward process, has been complicated by a weaker economy, massive global events like the COVID-19 pandemic, and possibly most notably, the rise of artificial intelligence in filtering out resumes. That last point is what the scope of this project was meant to be around, using a job-finder API alongside an LLM (in this case Gemini) to find job listings and adjust resume points to bypass the specific keywords that these ATS systems are looking for. To do so, I initially wanted to use the Adzuna API, a massive job-board, as well as the Gemini API, with the ability to upload a resume with pypdf and python-docx.

What ended up happening was that the Adzuna API was not functional for this project. While it has a trove of job information, it did not have the capacity to call on actual job descriptions that are needed to tailor a resume. Thus, I had to shift over to using the Google Jobs API, which instead looked up jobs listings on Google and pulled directly from the descriptions on there. Using this API proved far more useful, although another obstacle I faced was Gemini's free tier limiting how many calls I could use. This prevented me from testing extensively, as the app would simply not run due to those rate limits. Practically, this means the app is functional, but users will not be able to rapidly mass-apply to job listings in the way that I had originally sought out to accomplish.

If I were to do this project again, the main thing I would change is the LLM API used. Whether that be OpenAI or Anthropic, a paid, token-based LLM would be significantly faster and more effective than the rate-limited Gemini. While this would require a more in-depth structure requiring users to pay for tokens, or for them to have to host more locally with their own LLM API, the app would be far more functional than it is currently. After that, I would modify the

prompts embedded within the LLM API to be more built out and thorough so it can display job descriptions more succinctly and summarized better, as well as parse through listing information more effectively to find what jobs make the most sense for the uploaded resume, so that users have the “best match” immediately displayed in front of them.